

Introduction to Contract Theory and Auctions

The Envelope Theorem

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- We will study now a powerful result that appears in the theory of auctions, as well as in many areas of contract theory.
- A lot of the most important results in these areas can be derived as a consequence of the **Envelope Theorem**.
- The version of the theorem we will study in these slides is very simple, and makes a lot of strong assumptions.
- Nevertheless, this version will be sufficient for most of our purposes, since in particular, it applies to our study of symmetric independent private value auctions.

The envelope theorem: simple version

- Consider a function $f : X \times T \rightarrow \mathbb{R}$, where $X \subset \mathbb{R}^n$ is non-empty, compact, and convex, and $T \subset \mathbb{R}$ is an interval on the real line. Assume also that f is continuously differentiable in all of its arguments.
- The interpretation is that x is the choice variable, and t is a parameter.
- Define, for every t , the **upper envelope** or **value function** as

$$V(t) = \max_{x \in X} f(x, t).$$

- Denote by $x^*(t)$ the function that maps parameters into optimal choices, that is:

$$x^*(t) = \arg \max_{x \in X} f(x, t),$$

so that

$$V(t) = f(x^*(t), t).$$

- Suppose further that $x^*(t)$ is in the interior of X for every t and that $x^*(t)$ is differentiable.

The envelope theorem: simple version

- What is the effect of a change in the parameter t on $V(t)$?
- This question is of utmost importance because it yields useful comparative statics results.
- Comparative statics results are very useful in the analysis of economic models, because they tell us how the parameterized optimization problem changes when we change one of the exogenous parameters.
- Differentiation yields

$$V'(t) = f_1(x, t)|_{x=x^*(t)} \frac{dx^*(t)}{dt} + f_2(x, t)|_{x=x^*(t)} = f_2(x^*(t), t),$$

where the second equality follows because $x^*(t)$ is, by assumption (see the last bullet from the previous slide), it is an interior optimum which satisfies a first order condition:

$$f_1(x^*(t), t) = 0.$$

- Intuitively: the “ x effect” of a change in t is smoothed out because x is chosen optimally with respect to t . Hence, the only effect that remains of a change in t is the “ t effect”.

Envelope Theorem Illustration

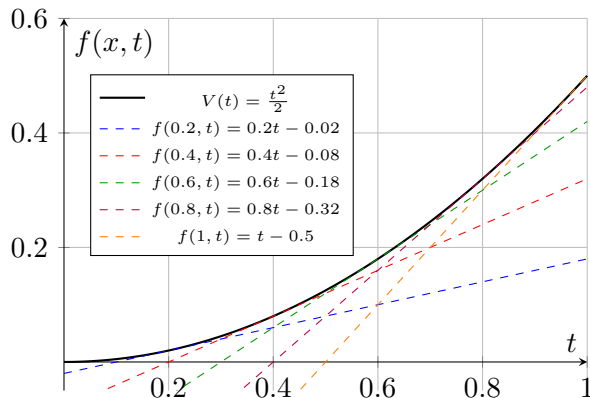
Consider $f(x, t) = xt - \frac{x^2}{2}$ and

$$V(t) = \max_{x \in [0,1]} \left\{ xt - \frac{x^2}{2} \right\}$$

- For fixed x , $f(x, t)$ is affine in t
- The maximizer is: $x^*(t) = t$
- Then

$$V(t) = f(x^*(t), t) = \frac{t^2}{2}$$

- $V(t)$ is the **upper envelope** of the collection $\{f(x, t)\}_{x \in [0,1]}$
- $V'(t) = t = f_2(x, t)|_{x=x^*(t)}$.



Using the envelope theorem to recover the value function

- By the envelope theorem we have that

$$V'(t) = f_2(x^*(t), t).$$

- Then, by the Fundamental Theorem of Calculus, we can recover $V(t)$ by integrating:

$$V(t) = V(t') + \int_{t'}^t f_2(x^*(s), s) ds = f(x^*(t'), t') + \int_{t'}^t f_2(x^*(s), s)$$

for some fixed $t' \in T$.

- In many applications, our parameter t can take on the value of zero, which we use as our fixed value t' in the expression above:

$$V(t) = V(0) + \int_0^t f_2(x^*(s), s) ds = f(x^*(0), 0) + \int_0^t f_2(x^*(s), s)$$

The Envelope Theorem in the Theory of Auctions

- We will now restate the simple version of the Envelope Theorem in the context of auctions.
- Consider a strategy bidding profile β_{-i} from bidders other than i , and suppose that bidder i , with value v_i , submits a bid of b_i .
- Then her expected payoff is denoted as $u_i(b_i, \beta_{-i}; v_i)$: under the symmetric private values model, this is equal to:

$$u_i(b_i, \beta_{-i}; v_i) = \Pr(\text{buyer } i \text{ wins the auction} \mid b_i, \beta_{-i}) \times v_i \\ - E[\text{buyer } i \text{'s payment to the seller} \mid b_i, \beta_{-i}]$$

- Notice that both the probability of winning and buyer i 's payment depend on the bid she submits, as well as β_{-i} , but not on her own private value v_i .
- Therefore, $u_i(b_i, \beta_{-i}; v_i)$ is affine¹ in v_i .

¹Linear: $f(v) = Av$. Affine: $f(v) = Av + b$. Please use grown-up terminology.

The Envelope Theorem in the Theory of Auctions

- Therefore, if β_{-i} is an equilibrium profile, at an equilibrium, buyer i must be playing a best response, i.e., a bid that maximizes her profit. Thus

$$V_i(v_i, \beta_{-i}) = \max_{b_i \geq 0} u_i(b_i, \beta_{-i}; v_i).$$

- We also have that:

$$b_i^*(v_i) = \arg \max_{b_i \geq 0} u_i(b_i, \beta_{-i}; v_i) = \text{BR}_i(\beta_{-i}; v_i)$$

so that:

$$V_i(v_i, \beta_{-i}) = u_i(b_i^*(v_i), \beta_{-i}; v_i)$$

- Notice that in the maximization problem above, the choice set (for bids) is the positive real line, which is not compact. However, this doesn't present a difficulty, since there always exists a bid $\bar{b}$ such that $u_i(b, \beta_{-i}; v_i) < 0$ for $b > \bar{b}$.

The Envelope Theorem in the Theory of Auctions

- If we further assume that u_i is continuously differentiable in all of its arguments, and that b_i^* is also differentiable and satisfies for each private value v_i :

$$\frac{\partial u_i}{\partial b_i}(b_i, \beta_{-i}; v_i) \big|_{b_i=b_i^*(v_i)} = 0,$$

then the conditions of the envelope theorem are met and we have that:

$$\frac{\partial V_i}{\partial v_i}(v_i, \beta_{-i}) = \frac{\partial u_i}{\partial v_i}(b_i^*(v_i), \beta_{-i}; v_i)$$

- Notice that in our symmetric private values model, this implies that:

$$\frac{\partial V_i}{\partial v_i}(v_i, \beta_{-i}) = \Pr(\text{buyer } i \text{ wins the auction} \mid b_i^*(v_i), \beta_{-i})$$

- Remark: the condition $\frac{\partial u_i}{\partial b_i}(b_i, \beta_{-i}; v_i) |_{b_i=b_i^*(v_i)} = 0$ holds if $0 < b_i^*(v_i) < \infty$ because then this is the first-order condition for a local maximum. It follows that it is necessary to check that it holds only if the maximum is at an extreme point, i.e., only if $b_i^*(v_i) = 0$, that is, it would be optimal to not bid at all. See also my comment on the domain of b_i above.

Application: A (slightly) quicker way to find the symmetric equilibrium in the FPA

- Suppose that the private values of the buyers are independent of each other and they fall on the interval $[0, \bar{v}]$, and that all bidders share the same cumulative distribution function F , i.e., this is the SIPV model.
- Assume there exists a monotonically increasing and differentiable symmetric equilibrium strategy β^* .
- Let β_{-i}^* be the strategy vector in which every buyer except i use β^* .
- Then the expected payoff of bidder i , when bidding b_i and her value is v_i is:

$$u_i(b_i, \beta_{-i}^*; v_i) = G_1^{(n-1)}((\beta^*)^{-1}(b_i)) \times (v_i - b_i).$$

- Since β^* is an equilibrium, then the best response of bidder i to the profile β_{-i}^* is to bid $\beta^*(v_i)$.
- Since β^* is differentiable, $(\beta^*)^{-1}$ is also differentiable which implies that $u_i(b_i, \beta_{-i}^*; v_i)$ is also differentiable.

Application: A (slightly) quicker way to find the symmetric equilibrium in the FPA

- Since β^* is monotonically increasing, and because $\beta^*(0) \geq 0$ and $\beta^*(\bar{v}) \leq \bar{v}$, it follows that $0 \leq \beta^*(v_i) \leq \bar{v}$ for all $v_i \in (0, \bar{v})$.
- From the previous point it follows that the first-order condition is met:

$$\frac{\partial u_i}{\partial b_i}(b_i, \beta_{-i}^*; v_i) \big|_{b_i = \beta^*(v_i)} = 0$$

- The envelope theorem implies that:

$$\frac{\partial V_i}{\partial v_i}(v_i, \beta_{-i}) = G_1^{(n-1)}((\beta^*)^{-1}(\beta^*(v_i))) = G_1^{(n-1)}(v_i).$$

- Note that $V_i(0, \beta_{-i}^*) = 0$. Integrating the above expression we get:

$$V_i(v_i, \beta_{-i}^*) = \int_0^{v_i} G_1^{(n-1)}(s) ds.$$

Application: A (slightly) quicker way to find the symmetric equilibrium in the FPA

- We also know that

$$V_i(v_i, \beta_{-i}^*) = G_1^{(n-1)}(v_i) \times (v_i - \beta^*(v_i))$$

- Equating the last two displayed equations yields:

$$G_1^{(n-1)}(v_i) \times (v_i - \beta^*(v_i)) = \int_0^{v_i} G_1^{(n-1)}(s) ds.$$

- Solving for $\beta^*(v_i)$ yields

$$\beta^*(v_i) = v_i - \frac{1}{G_1^{(n-1)}(v_i)} \int_0^{v_i} G_1^{(n-1)}(s) ds.$$

- The above is the same expression we get when we integrate by parts:

$$\beta^*(v_i) = \frac{1}{G_1^{(n-1)}(v_i)} \int_0^{v_i} s g_1^{(n-1)}(s) ds$$

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What have we obtained?

- We just saw another procedure to derive the equilibrium of the FPA in the SIPV model.
- What is the benefit of this procedure? We no longer need to check that β^* as derived above is an equilibrium because if a monotonically increasing, differentiable, and symmetric equilibrium exists it is necessarily given by the above expression, thanks to the Envelope Theorem.
- The envelope theorem can be used to derive the symmetric, monotonic, and differentiable equilibrium from other auction formats.
- The version of the theorem we saw here is very simple, and makes quite a bit of assumptions that are usually not satisfied in mechanism design or other contract theory applications.
- If you want to see a more powerful version, which requires less assumptions, please check the extra notes that I prepared.²

²In that note, I develop the Milgrom and Segal (2002) integral form of the envelope theorem. To read that note, I assume that you have a very good course in calculus and possibly also a course in Real Analysis. I recommend reading that note if you are either really curious, or mathematically/theoretically inclined.